

VU MANH HUNG

DATA ENGINEER

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Summary

Data Engineer with hands-on ownership of production analytics platforms and experience designing near-real-time advertising pipelines using Python, SQL, Airflow, dbt, BigQuery, and Google Cloud Storage. Practical experience in REST API, callback and file ingestion, incremental reprocessing, late-arriving data, deduplication, reconciliation, monitoring, and backfills. Translates reporting requirements into maintainable pipelines and validated datasets.

Professional Experience

Yuta Labs — Data Engineer

Apr 2026 – Present

Full-time

Near-real-time Campaign ROAS

- Owned the design and technical validation of a dual-path ROAS system combining cost data from Google, Meta, TikTok, and Unity with Adjust revenue; retained file-based ingestion as a reconciliation baseline while the callback path supported approximately 15-minute reporting freshness.
- Designed durable SQLite buffering for callback ingestion and batched publication to BigQuery, with rolling-window deduplication and reconciliation between file and callback revenue paths to investigate reporting discrepancies.
- Designed changed-hour ingestion using GCS shard-count watermarks to reprocess late-arriving files, and Airflow Dataset dependencies to publish shared cost data before downstream ROAS marts.

Analytics Platform Ownership & Redesign

- Took sole Data Engineering ownership of an inherited analytics platform; reconstructed data flows and reporting contracts across Airflow, BigQuery, and dbt while maintaining existing production workflows.
- Designed hourly GA4 ad-event reporting with a seven-day reprocessing window, keyed BigQuery MERGE, and reconciliation by app, country, and version to handle late updates and repeatable processing.
- Owned the integration design for Unity Ads cost data across three reporting pipelines and defined a staged Python refactor with shared cloud operations and injectable dependencies for offline replay and validation.

L'INNO — Data Engineer, Part-time

Mar 2025 – Aug 2026

Remote | Concurrent role

Blue Dragon Reporting Platform

- Maintained and enhanced the reporting platform, owning the existing GiveCloud flow and developing the PayPal reporting extension.
- Built a Python collector for four regional PayPal REST APIs (Vietnam, US, UK, Australia), landing daily data in Google Cloud Storage and loading the consolidated dataset into BigQuery.
- Handled duplicate transactions, refunds, and late-arriving updates; matched PayPal and GiveCloud records using email addresses and transaction timestamps.
- Designed fact and dimension models for transaction reporting, donor segmentation, recurring donations, and revenue trend analysis, with validation checks for data issues.
- Scheduled daily workflows with GitHub Actions, monitored runs, investigated failures, and performed historical backfills.
- Built Looker Studio dashboards for donation analysis and mapped donor and transaction data to Salesforce objects and fields.

Internal Analytics Data Pipeline

- Built ELT workflows for Timesheet, CRM, and ERP data; analyzed 1,600+ source tables in PostgreSQL and MariaDB to identify reporting entities and exclude metadata/system tables.
- Developed dbt models for a reporting-ready layer and orchestrated daily ingestion, transformation, and validation with Airflow; documented source-to-target mappings.

Accounting BI Dashboard

- Designed a Microsoft-based accounting BI solution using OneDrive files, Power Query, and Power BI; cleaned, standardized, and modeled financial data for expense tracking and operational monitoring.
- Built and maintained Power BI dashboards for financial reporting and business analysis.

Projects

Financial Market Data Collection & ELT Pipeline

Feb 2026 – Apr 2026

Personal Project

- Developed Python collectors using requests to retrieve and parse JSON data for OHLC prices, company listings, financial filings, and news sentiment from Polygon, Alpha Vantage, and SEC APIs.
- Deployed a single-node Hadoop cluster with HDFS and YARN on Google Compute Engine and configured Spark/PySpark for processing.
- Validated missing values, duplicates, API response status, and schema consistency before downstream processing.
- Designed a DuckDB dimensional warehouse with three fact tables and four shared dimensions for price, volume, company, and sentiment analysis.
- Implemented Spark transformations for cleansing and warehouse preparation, orchestrated end-to-end with Airflow DAGs.

Student Performance Prediction Pipeline

Jul 2025 – Sep 2025

iBME Lab, HUST

- Cleaned, standardized, and prepared approximately 1M student records for GPA/CPA and course-grade prediction; engineered features and built models with Python, Pandas, and Scikit-learn.
- Developed a Streamlit application for result visualization and exploratory analysis.

Education

Hanoi University of Science and Technology

Bachelor of Engineering in Electronics and Telecommunications Engineering
2020 – 2025 | Degree Classification: Good

Skills

Core: Python, SQL, Airflow, dbt, BigQuery

Cloud & Ingestion: Google Cloud Storage, REST APIs, callback and file ingestion

Data Engineering: ETL/ELT, workflow orchestration, incremental reprocessing, late-arriving data, reconciliation, monitoring, backfills

Data Modeling: Data warehouse, source-to-target mapping, fact and dimension modeling

Databases: PostgreSQL, MariaDB, DuckDB, SQLite

Supporting Tools: Git, GitHub Actions, Docker, Pandas, Spark/PySpark, dlt

BI: Looker Studio, Power BI, Metabase, Power Query

English: Aptis B1 (CEFR)